



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

PL: Aktywna alokacja ograniczonych danych rzeczywistych w pętli rzeczywistość–symulacja–rzeczywistość: budowa cyfrowych bliźniaków, uczenie w nich i przenoszenie wyników do rzeczywistości

EN: Active allocation of limited real data in the real-to-sim-to-real loop: building digital twins, learning in them and transferring the results back to reality

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:



- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,
- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	<i>(Oct 2025 – Feb 2026)</i> Research target: define the problem and map the state of the art. T1.1 Systematic literature review in four areas: (a) sim-to-real transfer and domain randomization, (b) learning under distribution shift and domain adaptation, (c) neural scene reconstruction (NeRF, 3D Gaussian Splatting), (d) learning-based visual navigation. <i>D:</i> annotated bibliography and a written research-gap statement (basis of §6). T1.2 Identification of the open problem: how much real data sim-to-real transfer needs, and whether reconstructed scenes can replace real data. <i>D:</i> problem statement and preliminary research questions, discussed with the supervisor. T1.3 Doctoral School courses of semester 1. <i>D:</i> completed courses. Place: K46.



Semestr 2	<i>(Mar – Sep 2026)</i> Research target: formulate the research plan and select the experimental basis. T2.1 Formulation of the data-efficient real-to-sim-to-real thesis, research questions RQ1–RQ4 (capture less, learn robustly in an imperfect twin, few real data, budget of the whole loop) and hypotheses H1–H4 with measurable decision rules and evaluation tiers A–C. <i>D:</i> §7 and §9 of this IPB. T2.2 Comparative survey of public benchmarks and datasets (e.g. ScanNet++, HM3D), simulators (e.g. Habitat-Sim) and 3D Gaussian Splatting implementations against the needs of RQ1–RQ4 (real video + reference scan, licence, compute cost). <i>D:</i> tool-selection note with a shortlist for Stage I. T2.3 Individual Research Plan agreed with the supervisor. <i>D:</i> this IPB, signed and submitted by 30.09.2026. Place: K46.
Semestr 3	<i>(Oct 2026 – Feb 2027)</i> Stage I – capture less (RQ1, H1). T3.1 Tier A set-up in both testbeds (§9): navigation, ≥ 20 public scenes (10 held out) with laser-scan references and separate phone captures; manipulation, ≥ 20 simulator task–object configurations (10 held out) with held-out physical parameters. Existing 3DGS pipeline and system identification from the T2.2 shortlist; twins at ≥ 4 capture budgets; pilot fixing task difficulty (below the P ceiling) and the operator-time cost model; imitation-learned policies in the twins; pre-registration of the protocol (§7, §9). <i>D:</i> working loop and protocol in the repository for both testbeds, by Jan 2027. Place: K46, WCSS. T3.2 Validation stand: written agreement on robot access (planned cooperation with the K29 Denali laboratory for the mobile robot; a PWR manipulator if available) and sensor decision (SzD Minigrant application if a sensor must be bought). <i>D:</i> agreement, by Nov 2026. T3.3 Task-aware, uncertainty-guided capture of views and interactions vs. uniform, task-blind and risk-weighted selection. <i>D:</i> first budget–success curves (first H1 result), by Feb 2027.
Semestr 4	<i>(Mar – Sep 2027)</i> Stage I completed; Stage II – learn robustly in an imperfect twin (RQ1–RQ2, H1–H2). T4.1 Uncertainty-aware randomization (appearance, geometry, physical parameters) and representation-based weighting of twin samples vs. uniform domain randomization at an equal capture budget, in both testbeds. <i>D:</i> H1 completed and first H2 result on tier A, by Apr 2027; H2 completed, by Jul 2027. Place: K46, WCSS. T4.2 Prepare and submit article P1 to NeurIPS 2027 (RQ1–RQ2: task-aware capture, H1, and first uncertainty-aware learning results, H2, in manipulation and navigation). <i>D:</i> submission, May 2027 (§11). T4.3 Stage III set-up: trial-selection and correction code on tier A (Jul–Sep 2027). <i>D:</i> first H3 result, by Sep 2027. T4.4 Grants and international activity: NCN PRELUDIUM application; poster at an international summer school; foreign co-author sought for P2; NAWA Bekker application for the sem. 6 visit. <i>D:</i> applications, poster.



Semestr 5	(Oct 2027 – Feb 2028) Mid-term evaluation (Nov 2027): autoreferat (≤ 5 pages, ~mid-Oct 2027) and 15-min presentation. Stage III – collect few real data (RQ3, H3). T5.1 Active selection of real trials (predicted gap, uncertainty) vs. random selection; correction of the twin (appearance and physical parameters) and fine-tuning of the policy with the selected trials, in both testbeds. D: H3 completed on tier A, by Jan 2028. Place: K46, WCSS. T5.2 First robot validation campaign (tier C, 2 PWR settings, ~8 robot-hours): own captures, twins, uniform capture vs. the loop. D: validation report. Place: Denali (and the manipulator's lab, if agreed). Without robot access, the campaign moves to sem. 7; H1–H3 are decided on tier A either way. T5.3 Article P2 (active selection of few real data and twin correction, RQ3) submitted to ICML 2028 (~late Jan 2028; CVPR 2028, ~Nov 2027, only if H3 is complete by then). If P1 was rejected, its revision goes to CVPR 2028. D: submission(s).
Semestr 6	(Mar – Sep 2028) Stage IV – budget of the whole loop in both testbeds (RQ4, H4). T6.1 Real-data budget curves of the full loop vs. real-only learning ("exchange rate") on tier A in manipulation and navigation, with the allocation methods unchanged, plus the tier-B check against published paired sim/real evaluations. D: first-testbed result by Apr 2028; H4 completed, by Sep 2028. Place: K46, WCSS. T6.2 Foreign research visit (3 months, NAWA Bekker; if not funded, a 5–30-day Erasmus+ short-term mobility to a European lab; hosts in §12): joint experiments. D: visit report. T6.3 Article P3 (budget law of the full loop across both testbeds) submitted to NeurIPS 2028 (~May 2028); if not ready, to ICLR 2029 (~late Sep 2028). D: submission.
Semestr 7	(Oct 2028 – Feb 2029) Consolidation (RQ1–RQ4). T7.1 Second robot validation campaign (tier C) of the full loop; final analysis of H1–H4 across tiers and testbeds. D: final results. Place: K46, Denali. T7.2 Release of code and experiment scripts. D: public repository. T7.3 Optional article P4 (consolidated study) to ICLR 2029 (~late Sep 2028) or CVPR 2029 (~Nov 2028); start of dissertation writing. D: submission (optional), dissertation outline.
Semestr 8	(Mar – Sep 2029) T8.1 Editing of the doctoral dissertation; final version. D: dissertation submitted by 30.09.2029 (§4). T8.2 Responses to reviews and camera-ready versions of pending articles. D: revised manuscripts.

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.



30.09.2029

(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Modern machine learning is limited less by models than by **data in the target domain**. Systems that act in or measure the physical world, such as robot manipulators, mobile robots and other autonomous or perception systems, need large amounts of experience, and in the real world that data is slow, expensive and sometimes risky to collect. A policy or model is therefore often trained in simulation and then used on real data, where it often performs worse (the **sim-to-real gap**).

Digital twins built from real data narrow this gap. Neural scene reconstruction (3D Gaussian Splatting, Neural Radiance Fields) turns a short real capture into a photorealistic model of the geometry and appearance of a scene, and system identification estimates its physical and dynamic parameters (masses, friction, actuation) from a few real interactions. This gives a **real-to-sim-to-real loop**: collect real data, build a twin, learn in the twin, check and correct with a little more real data, deploy. Building twins has quickly become routine, with dozens of papers a year and more than ten open-source twin simulators. What remains open is the **real data** the loop consumes. Each step spends it: the capture that builds the twin, the real data used to make learning robust to what the twin got wrong, and the real-world trials or interactions used to check and correct the result. Today these amounts are chosen by habit: data is collected uniformly, learning uses generic randomization, and real trials are picked at random or by hand. How much real data a twin actually saves has, to our knowledge, not been measured systematically across tasks.

What the dissertation proposes. The real data needed by the loop can be reduced by **allocating it actively**: collect only the data the task needs to build the twin, learn so that the result is robust to the twin's errors, and collect only the few real-world data or interactions that close the remaining gap. Two signals, both computed from data the loop already collects, guide these choices: the **uncertainty of the twin** (where its reconstruction or its identified parameters are unreliable) and the **learned model's own internal representations** (where twin and real inputs look different to it). The result is measured as a budget–performance curve: how much real data the loop needs to reach a given real task performance, compared with uniform allocation and with learning from real data only.

Why this topic, and why in this discipline. The object is a general, task- and domain-agnostic data-efficient learning methodology (active learning, learning under distribution shift, uncertainty estimation), not a particular robot, task, simulator or benchmark, which places it in *information and communication technology*. The student uses existing twin pipelines, simulators and public datasets and does not develop new ones. The methods are tested with the same settings on two testbeds of equal status, **robotic manipulation** and **visual navigation**, where public data make it possible to count real data exactly without an own robot. Trials on a real robot (planned cooperation with a PWr robotics laboratory) validate, but do not decide, the conclusions. The topic fits the Department of Artificial Intelligence (K46), whose research groups include representation learning. The computation runs on the GPU infrastructure available to PWr researchers (WCSS, PLGrid).



Potential application areas: faster and cheaper deployment of learned systems in new places, such as robot workcells, warehouses, buildings and inspection sites, with less time spent collecting data and supervising real trials; guidance on how much real data to collect for a twin; and, beyond robotics, any model trained on reconstructed or simulated data and used on real sensor data (industrial digital twins, perception for autonomous systems), where real data must be collected on a budget.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Real-to-sim-to-real with digital twins. A real-to-sim-to-real loop builds a simulation of the target scene or object (a *digital twin*) from real data, learns a policy or model in it and transfers it back. Generic simulators such as Habitat [1] train at scale, but not in the target scene. 3D Gaussian Splatting (3DGS) [2] turns a short camera capture into a real-time, photorealistic twin of geometry and appearance, and policies trained in such twins transfer in manipulation (RialTo [3], SplatSim [4]) and in navigation and locomotion (EmbodiedSplat [5], Vid2Sim [6], GaussGym [7]); GPU simulators ship twin environments (ManiSkill3 [8]). The physical side is identified from real data: BayesSim [9] infers a posterior over simulator parameters from real trajectories, and PhysTwin [10] recovers the geometry, physical properties and appearance of deformable objects from sparse videos. Building a twin is no longer the bottleneck; **real data** is: the capture that builds the twin and the real-world data or interactions that correct it and the policy. Current systems fix this cost by hand: EmbodiedSplat uses a 20–30-minute phone capture per scene, and RialTo's ablation over 0–15 real demonstrations is the closest published analysis of a real-data budget.

How much real data does transfer need? Real-only scaling studies relate performance to the number of real demonstrations in manipulation [11]; in navigation, diversity matters more than quantity and more data from a known location saturates quickly [12]. On the twin side, CASHIER [13] reports performance that scales super-linearly with human effort, X-Sim [14] matches behaviour cloning with ten times less data-collection time, and sim-and-real co-training raises real success by 38% on average [15]. Lower-fidelity simulation can transfer better [16], so more capture is not automatically better. All of these report results at fixed or hand-chosen budgets in one domain. **We found no study that varies the real-data budget of a twin as an experimental variable, or that fits real-data budget curves of the whole loop against real-only learning across domains.**

Capturing less: active reconstruction and active identification. Active view selection chooses the images that improve a reconstruction most: FisherRF [17] maximizes the expected information gain on radiance-field parameters, GenNBV [18] learns a next-best-view policy that generalizes across scenes, and Bayes' Rays [19] estimates a volumetric uncertainty field for a trained neural radiance field. Risk-aware view acquisition [20] weights FisherRF by safety-critical regions during exploration. For physical parameters, ASID [21] uses an initial simulator to design exploration that collects a small amount of informative real data for identifying articulation, mass and other parameters. Their objective is reconstruction quality, safe exploration or parameter accuracy, not the success of a policy trained in the twin; we found at most two task-aware capture papers a year in 2023–2026. Public data allow budgets to be varied offline: ScanNet++ [22] pairs laser scans with DSLR images and a separate phone stream.

Learning robustly in an imperfect twin. Domain randomization varies appearance [23] or dynamics [24] uniformly within hand-set ranges, and SimOpt [25] adapts the randomization distribution from a



few real rollouts. Domain adaptation aligns sim and real features in co-training [26]. Frozen encoders (DINOv2 [27]) make twin and real samples comparable, and Phys2Real [28] trains with uncertainty over physical parameters. Still, a twin's own **reconstruction uncertainty** is rarely used as a training signal, and weighting twin samples by representation distance to a small real set appeared in one paper in 2023–2026.

Collecting few real data. Active learning selects the most informative real samples for sim-to-real adaptation in grasping [29], AMF [30] chooses which tasks to demonstrate under a budget, and active experiment selection reduces the cost of real evaluation [31]. Prediction-powered inference [32] combines many cheap predictions with a few labels, SureSim [33] applies it to simulated and real trials, and SIMPLER [34] shows, on paired simulated and real evaluations, that simulation can track real performance of manipulation policies. GaussTwin [35] corrects a 3DGS twin from photometric error. They select demonstrations, tasks or trials, not the real data that should correct a twin and its policy.

Research gap. Real-to-sim-to-real works, but its real-data cost is set by hand and spent uniformly, and each study measures it in one domain. Missing are (i) task-aware, uncertainty-guided capture of both appearance and physical parameters, judged by downstream success rather than reconstruction or parameter accuracy (RQ1), (ii) learning driven by the twin's uncertainty and by representation distance to a small real set (RQ2), (iii) active selection of the few real data that correct the twin and the policy (RQ3), and (iv) a total real-data budget curve of the full loop against real-only learning, measured the same way in manipulation and navigation (RQ4).

References

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7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Thesis. The real data needed in a real-to-sim-to-real loop can be reduced substantially by **allocating it actively**: collect only the data the task needs to build the twin, learn so that the policy or model is robust to what the twin got wrong, and collect only the few real-world data that close the remaining gap. The twin's uncertainty and the model's own representations guide all three steps. Testbeds (equal status, same settings): robotic manipulation and visual navigation.

Tier A protocol (proxy reality). In each testbed a **reference from a separate, higher-fidelity source plays "reality"**, and the *twin* (neural reconstruction plus system identification) is built from a subset of a separate, cheaper capture, so the two share no data. Navigation: ≥ 20 public indoor scenes (10 held out), laser-scan and DSLR reference, twin from the phone capture. Manipulation: a physics simulator with held-out physical parameters and its own rendering, the weaker proxy, checked by tier B (published paired sim/real evaluations of real policies). Real data: capture = views and interaction samples given to the twin, real trials and demonstrations = episodes in the reference, summed in one operator-time cost. Metric P: task success rate in the reference. Tier A decides H1–H3 in both testbeds; tiers B and C (real robot) report agreement. Tests: one-sided, $\alpha = 0.05$, scene/seed bootstrap, Holm-corrected per hypothesis, pre-registered.

RQ1 (real $\rightarrow$ sim: capture less). How little real data does a useful twin need, and can the task guide it?

H1. Task-aware, uncertainty-guided capture (next data where task-relevant parts of the twin are most uncertain) reaches the P of uniform capture with $\geq 40\%$ less capture. *Decision:* on fitted capture–P curves, $C_{\text{task}}(\tau_1) \leq 0.6 \cdot C_{\text{uniform}}(\tau_1)$, and the upper 95% bound of $C_{\text{task}} / C_{\text{recon}}$ (task-blind uncertainty selection) is < 1 ; $\tau_1 = P$ of uniform capture at its largest budget.

RQ2 (in sim: learn robustly in an imperfect twin). How should learning in the twin be made robust to its errors?

H2. Uncertainty-aware training (augmentation and sample weights from the twin's uncertainty and from the representation distance to a few held-out real samples) improves transfer over uniform domain randomization at an equal capture budget. *Decision:* mean paired P gain ≥ 10 pp on held-out scenes, lower 95% bound > 0 , at each of ≥ 2 capture budgets.

RQ3 (sim $\rightarrow$ real: few real data). Which few real-world data close the remaining gap, and how should they correct twin and model?

H3. Actively selected real trials (by predicted gap or uncertainty), used to correct twin and model, reach the target P with $\geq 50\%$ fewer than random selection. *Decision:* $N_{\text{active}}(\tau_3) \leq 0.5 \cdot N_{\text{random}}(\tau_3)$ on fitted trial–P curves, upper 95% bound of the ratio < 1 ; $\tau_3 = P$ of random selection at its largest budget.

RQ4 (whole loop: budget and generality). What real-data budget does the full loop need versus real-only learning, across domains?

H4. (a) In each testbed the full loop (H1–H3) reaches the target P with $\leq 10\%$ of the real data needed by real-only learning (imitation of demonstrations in the reference, same encoder and initialization). *Decision:* $B_{\text{loop}}(\tau_4) \leq 0.1 \cdot B_{\text{real}}(\tau_4)$, upper 95% bound of the ratio < 0.2 ; $\tau_4 = P$ of real-only learning at its largest budget; also reported: a uniform loop and a world-model simulator. (b) This holds in



both testbeds with the allocation methods and their hyperparameters unchanged, the H1–H3 effects keep their sign, and tier B agrees in direction.

Decision rule. A hypothesis holds when all its parts pass; a failed one is reported as a measured budget curve and does not block the next step.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The dissertation is carried out in **information and communication technology** (*informatyka techniczna i telekomunikacja*), in the area of machine learning. Its object is the **real-to-sim-to-real learning loop** in general, task- and domain-agnostic: building a digital twin (appearance, geometry and, where relevant, physical parameters) from limited real data, learning a policy or model in it and transferring the result back to reality, and how to make this loop work with **less real data**. Building twins, simulators or benchmarks, control design and mechanical engineering are not the object of the research; the student uses existing tools and public data.

Key original contribution. To our knowledge, the first treatment of the real data in the whole real-to-sim-to-real loop as **one budget to be allocated actively**, with a method for each step of the loop and real-data budget curves measured with the same methods in two domains, robotic manipulation and visual navigation.

Contributions to machine learning methodology:

1. **Capture less (RQ1, H1).** A task-aware capture method that chooses which real data to collect for a twin (views for its appearance and geometry, interactions for its physical parameters) from the twin's uncertainty in the parts that matter for the task, instead of for reconstruction quality alone, and a test of how much capture it saves against uniform and task-blind selection.
2. **Learn robustly in an imperfect twin (RQ2, H2).** An uncertainty-aware training method that turns the twin's per-region and per-parameter uncertainty and the representation distance to a small set of real samples into augmentation and sample weights, compared with uniform domain randomization at an equal capture budget.
3. **Transfer with few real data (RQ3, H3).** An active selection method for real-world trials or interactions, driven by the predicted twin-to-real gap and its uncertainty, and a procedure that uses them to correct both the twin and the learned model, compared with random selection.
4. **Budget curves for the loop (RQ4, H4).** Budget–performance curves of the full loop that give the "exchange rate" between twin and real data (how much real data the loop needs, relative to learning from real data only, to reach a given real task performance), and a test of whether the methods hold unchanged in both domains.

Contributions in engineering terms (ITiT):

5. **Evaluation protocol and software (supports all RQs).** A proxy-reality protocol on public data that counts real data exactly without a robot (a reference from a separate, higher-fidelity source as "reality", a low-budget twin from a separate capture as the simulator), applied in each domain, and open-source code for the three methods and the budget curves, built on existing twin pipelines and simulators. It tells practitioners roughly how much real data to collect for a new task or site.

Significance for the discipline. Real target-domain data is the main cost of deploying learned systems. Methods that decide where it is worth spending, and a measured estimate of how much of it a twin



can replace, are relevant to active learning, learning under distribution shift and uncertainty estimation in general, beyond robotics.

Dissemination. Results are planned for peer-reviewed conferences worth 200 points and assigned to ITiT on the ministerial list of 5.01.2024: NeurIPS, ICML and ICLR (machine learning) and CVPR (computer vision), with ICCV and ECCV for resubmissions. P1 (RQ1–RQ2, task-aware capture and first uncertainty-aware training results) targets NeurIPS 2027; P2 (RQ3, active selection of real data and twin correction) ICML 2028 or CVPR 2028; P3 (RQ4, budget curves in both domains) NeurIPS 2028, with ICLR 2029 as the fallback.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The research follows the loop: real data → twin → learning in the twin → a few real trials → correction of the twin and the policy → deployment. Four stages (I–IV) match §3 and the research questions of §7: capture less (RQ1), learn robustly in an imperfect twin (RQ2), collect few real data (RQ3), and measure the budget of the whole loop (RQ4). The methods are domain-agnostic: they act on the twin's uncertainty, on task relevance measured in the twin and on learned representations. The twin has two parts: geometry and appearance, reconstructed with existing open-source 3D Gaussian Splatting (3DGS) [2] pipelines, and physical or dynamic parameters, identified from real interactions as a posterior (BayesSim-type [9]). No simulator or benchmark is developed. Robotic manipulation and visual navigation are equal testbeds with the same protocol.

Tier A protocol: proxy reality (decides the hypotheses). In each testbed, "reality" is a reference from a separate, higher-fidelity source, and the twin is built from a subset of a separate, cheaper capture, so twin and reference share no data. *Navigation:* ScanNet++ [22] gives each scene a laser scan, DSLR images and a separate phone RGB-D stream. On ≥ 20 scenes (10 held out from all tuning), the reference is the laser-scan mesh textured from the DSLR images and rendered in Habitat [1] with a robot-camera model; the twin is 3DGS from a subset of the phone stream, run as a mesh (as in EmbodiedSplat [5]) or with a 3DGS renderer (GaussGym [7]); its physical part is the robot's actuation noise. Tasks: image-goal and point-goal navigation with a minimum geodesic distance set in a pilot so that P stays below its ceiling. *Manipulation:* ManiSkill3 [8] tasks (≥ 20 task–object configurations, 10 held out), where reality is the simulator with held-out ground-truth physical parameters (mass, friction, articulation) and its own rendering, and the twin is 3DGS from a subset of its views plus parameters identified from a few recorded interactions. Physics engine and object models are shared, so this is the weaker proxy (unmodelled dynamics are not tested); tier B checks it. *Both:* real data is counted exactly: capture = views and interaction samples given to the twin, real trials and demonstrations = episodes in the reference, summed in one operator-time cost at a pre-registered rate; the largest budget stays below the full capture. In navigation the proxy's error to held-out real images is reported (PSNR, LPIPS, depth) and each hypothesis is re-checked for sign with a second reference (3DGS from all DSLR images). *Tier B:* the twin must rank manipulation policies as published paired sim/real evaluations do (SIMPLER [34]). *Compute:* WCSS Lem (NVIDIA H100), PLGrid (to be applied for). Policies are small heads on frozen visual encoders, trained mainly by imitation of a privileged expert in the twin (shortest-path planner; motion-planning expert), with reinforcement learning on a subset. P = task success rate.

Stage I – task-aware capture (RQ1, H1; sem. 3–4).

- *Method:* the twin's uncertainty, per region (Fisher information [17] or a Bayes' Rays-type [19] field, adapted to 3DGS) and per physical parameter (posterior spread), is weighted by **task relevance** measured in the current twin without training: how often expert trajectories pass through a region, and how much the expert's success changes across a parameter's posterior. The next views or interactions are chosen greedily where task-weighted uncertainty is highest until the budget is



spent. Candidates come from the recorded pool; capture time is also estimated as a tour of the chosen views.

- *Baselines*: uniform capture, task-blind uncertainty selection (FisherRF-type next-best-view [17]; ASID-type exploration for parameter accuracy [21]) and risk-weighted view selection [20], at the same budget.
- *Metric*: budget–P curves on tier A with ≥ 4 capture budgets per testbed; capture needed to reach the P that uniform capture reaches at the largest budget.
- *Success criterion (H1)*: $\geq 40\%$ less capture than uniform at equal P, and less than task-blind selection (upper 95% bound of the ratio < 1), as in §7.

Stage II – uncertainty-aware learning in an imperfect twin (RQ2, H2; sem. 4).

- *Method*: (a) randomization whose strength follows the twin's uncertainty: appearance and geometry perturbations per region, and physical parameters sampled from their identified posterior instead of hand-set ranges; (b) down-weighting twin samples that lie far from a few real samples in frozen DINOv2 [27] feature space. The real samples are a held-out slice of the capture, not used to fit the twin, and count towards the budget.
- *Baselines*: uniform domain randomization of appearance [23] and dynamics [24] with tuned ranges, no randomization, and feature alignment [26] as an extra comparator; all at the same capture budget.
- *Metric and criterion (H2)*: tier-A P on held-out scenes, paired by scene; ≥ 10 pp higher P than uniform domain randomization at an equal capture budget (lower 95% bound > 0), at each of ≥ 2 budgets.

Stage III – few real trials and correction (RQ3, H3; end of sem. 4 – sem. 5).

- *Selection*: candidate real trials (start–goal pairs, object configurations) are scored by the predicted twin-to-real gap: disagreement of a small policy ensemble plus the twin's uncertainty along the planned trajectory. The top-k are executed in reality (tier A: the reference).
- *Use*: the trials (i) correct the twin, by re-capturing or re-weighting regions where real and twin observations disagree (photometric correction as in [35]; re-captured views are counted) and by updating the physical-parameter posterior [9, 25]; and (ii) fine-tune the policy by co-training with the real trials [15]. Real P is estimated from few trials with prediction-powered inference [32, 33].
- *Baselines*: random selection and uniform coverage of the same number of trials; selection rules [29–31] where they apply.
- *Success criterion (H3)*: the target P is reached with $\geq 50\%$ fewer real trials than random selection (upper 95% bound of the ratio < 1).

Stage IV – budget of the whole loop in both testbeds (RQ4, H4; sem. 6).

- *Budget curve*: total real data (capture + trials, one operator-time cost) against P for the full loop (Stages I–III) and for **real-only learning**: imitation of expert demonstrations collected in the reference with the same frozen encoder and initialization (real-only reinforcement learning also shown). The "exchange rate" is the ratio of the budgets that reach the target P; real-only scaling results [11, 12] suggest the curve shapes. The curve is also reported for a uniform loop and for a released pretrained world model as the simulator (not trained).
- *Success criterion (H4)*: with the three allocation methods and their hyperparameters unchanged (the learner is each task's standard one), in each testbed the loop reaches the target P with $\leq 10\%$ of the real data needed by real-only learning (upper 95% bound of the ratio < 0.2), the H1–H3 effects keep their sign, and tier B agrees in direction.

Tier C validation. Two robot campaigns (sem. 5 and 7) with own phone or RGB-D captures at PWr: a mobile robot in 2 rooms and, where access is agreed, a tabletop manipulator. Uniform capture versus the loop, plus the few selected trials; about 8 robot-hours per campaign (3 policies $\times$ 2 settings $\times$ 30 episodes at ~ 2 min, plus 2×20 trials). Tier C reports agreement only and never tunes thresholds.

Statistics and reproducibility. Tests follow §7; the protocol is pre-registered before each stage and changes are logged. Code is released where the dataset licences permit.



10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Systemy sztucznej inteligencji, które działają w prawdziwym świecie, na przykład roboty chwytające przedmioty lub poruszające się po budynkach, uczą się na bardzo wielu przykładach i próbach. W rzeczywistości zbieranie takich danych jest powolne i kosztowne, dlatego systemy te często ćwiczą w symulacji komputerowej. Coraz częściej jest nią cyfrowy bliźniak: wierna kopia prawdziwego miejsca lub urządzenia, którą można odtworzyć z nagrań i pomiarów, łącznie z wyglądem, kształtem i tym, jak rzeczy się poruszają. Powstaje w ten sposób pętla: zbieramy dane w rzeczywistości, budujemy kopię, uczymy w niej system, sprawdzamy go w rzeczywistości, poprawiamy i dopiero wtedy wdrażamy. Każdy z tych kroków wymaga jednak prawdziwych danych, a ich zbieranie jest kosztowne.

Celem rozprawy jest sprawdzenie, jak przejść przez tę pętlę z jak najmniejszą ilością prawdziwych danych, niezależnie od zadania. Zamiast zbierać wszystko po równo, będziemy zbierać przede wszystkim te dane, które są ważne dla zadania i o których kopia wie jeszcze za mało. Ponieważ kopia nigdy nie jest idealna, nauczymy system tak, aby nie polegał na jej niepewnych fragmentach i radził sobie z jej błędami. Na koniec wybierzemy tylko te nieliczne próby w rzeczywistości, które najwięcej mówią o tym, gdzie system jeszcze zawodzi, i wykorzystamy je do poprawienia zarówno kopii, jak i systemu. Wskazówek dostarczy sama kopia, która wie, gdzie jest niepewna, oraz sposób, w jaki uczony system opisuje to, co widzi. Te same metody sprawdzimy na dwóch różnych zadaniach: chwytaniu przedmiotów ramieniem robota i poruszaniu się robota po budynkach. Badania będą prowadzone przede wszystkim na publicznie dostępnych danych, a wybrane wyniki zostaną potwierdzone na prawdziwym robocie.

Spodziewanym efektem są metody, które pozwolą zbierać mniej danych, uczyć skuteczniej i wykonywać mniej kosztownych prób w rzeczywistości, oraz pomiar, ile prawdziwych danych potrzeba, gdy system uczy się w cyfrowej kopii. Może to przyspieszyć i potęgować wdrażanie uczących się systemów w fabrykach, magazynach, szpitalach i biurach. Opracowane oprogramowanie zostanie udostępnione publicznie.

Abstract for general public

Artificial intelligence systems that act in the real world, for example robots that grasp objects or move around buildings, learn from a very large number of examples and trials. In reality, collecting such data is slow and costly, so these systems often practise in a computer simulation. Increasingly, this is a digital twin: a faithful copy of a real place or device that can be rebuilt from recordings and measurements, including how things look, their shape and how they move. This creates a loop: we collect data in reality, build the copy, train the system in it, check it in reality, correct it and only then deploy it. Each of these steps, however, needs real data, and collecting it is costly.

The goal of this dissertation is to find out how to go through this loop with as little real data as possible, whatever the task. Instead of collecting everything evenly, we will collect mainly the data that matters for the task and about which the copy still knows too little. Because the copy is never perfect, we will train the system so that it does not rely on the uncertain parts of the copy and copes with its errors. Finally, we will choose only the few trials in reality that tell the most about where the system still fails, and use them to correct both the copy and the system. Guidance will come from the



copy itself, which knows where it is uncertain, and from the way the learning system describes what it sees. We will test the same methods on two different tasks: grasping objects with a robot arm and a robot moving around buildings. The research will be carried out mainly on publicly available data, and selected results will be confirmed on a real robot.

The expected result is methods that make it possible to collect less data, train more effectively and carry out fewer costly trials in reality, and a measurement of how much real data is needed when a system learns in a digital copy. This may make deploying learning systems in factories, warehouses, hospitals and offices faster and cheaper. The software developed will be made publicly available.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

May 2027 / maj 2027

Paper **P1** submitted for review to the main track of the **Conference on Neural Information Processing Systems (NeurIPS 2027)**. NeurIPS is peer-reviewed international conference proceedings worth **200 points** on the ministerial list (conference Lp. 87), assigned to the discipline *informatyka techniczna i telekomunikacja*. P1 reports Stage I and the first Stage II results (RQ1–RQ2): task-aware, uncertainty-guided capture of real data for digital twins (H1) and uncertainty-aware learning in imperfect twins (first H2 results), evaluated in manipulation and navigation by the real data they save (task T4.2). If it is rejected, the revised paper goes to **CVPR 2028** (expected deadline November 2027, also 200 points), and then to the next 200-point conference (§12).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

Planned outputs by year (200-point conferences assigned to the discipline, list of 5.01.2024).

Year	Papers (planned submission, 200 points each)	Code	Grants	Mobility, events
1 (2025/26)	—	—	—	—



2 (2026/27)	P1 (RQ1–RQ2, task-aware capture, first H2 results): NeurIPS 2027, May 2027 (§11)	Loop on tier A	SzD Minigrant; NCN PRELUDIUM; NAWA Bekker	Summer school poster
3 (2027/28)	P2 (RQ3, few real data, twin correction): ICML or CVPR 2028; P3 (RQ4, budget of the full loop in both testbeds): NeurIPS 2028	Trial selection	—	Mid-term; 3-month visit (sem. 6)
4 (2028/29)	Optional P4, consolidated study: ICLR 2029 or CVPR 2029	Full release	—	Dissertation

P1–P3 are the linked core of the dissertation; P4 is optional.

Foreign research visit (candidate hosts, not yet contacted). (1) Z. Kira's group, Georgia Tech: EmbodiedSplat [5]. (2) Multi-robot Systems Lab, Stanford University (M. Schwager): Gaussian-splatting scene models. A European host (Erasmus+ route) working on active reconstruction or sim-to-real transfer, and a foreign co-author for P2, are sought with the supervisor in sem. 4.

Collaboration at PWr. Robot validation is planned with the K29 *Denali* laboratory (Pioneer 3-DX, Jaguar 4x4, ROS 2) and, if available, a PWr manipulator; a written agreement is task T3.2 (Nov 2026).

Risks and mitigation (semester affected).

- *Task-aware capture gives no gain* (3–4): the measured capture–P curves still answer RQ1, and Stages II–III have their own baselines (§7).
- *Twin uncertainty is poorly calibrated* (3–4): two estimators and an ensemble fallback, checked against the reference.
- *Proxy reality is easier than reality* (4–7): separate sources for twin and reference; tiers B and C check the direction (§9).
- *Real-only learning is strong at small budgets* (6): gains saturate quickly in a known location [12]; H4 is reported as a full budget curve.
- *Scooping* (3–6): twin-based robot learning moves fast [3–7, 13]; early preprints, monthly monitoring.
- *Dataset licence or robot access* (3, 5): other public scans with reference geometry, own phone captures of PWr rooms; without a robot, tier C moves by a semester and hypotheses are still decided on tier A.
- *Rejection at a top venue* (4–7): each paper has a resubmission path inside the 200-point set 2–4 months later (NeurIPS → CVPR → ICML → ECCV/ICCV).

Prior work (background only). A pre-PhD NLP paper (ACL 2025 SRW, forecasting from hidden states); not part of the dissertation.

Ethics and data. Own captures are anonymized (GDPR); public datasets are used under their licences.

13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.



- **Weekly meeting** (~1 h, in person at K46 or online via MS Teams) to discuss progress, results and next steps.
- **Monthly written progress summary** (completed tasks, results, problems, plan). It lists every IPB task from §3 with its **% completion**, in the same form as the task table of the mid-term self-report.
- **Semester review** against the IPB schedule (§3), before each annual report and the mid-term evaluation. Deviations from the plan and their reasons are recorded and agreed with the supervisor.
- **Supervisor co-signs** the annual reports and the mid-term self-report (autoreferat).
- **Paper-writing sprints** before each submission deadline, with joint editing in Overleaf. The supervisor co-authors and reviews the papers before submission.
- **Ongoing communication** via e-mail / Teams chat. The code and experiment logs are in a shared Git repository that the supervisor can access.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor